

## PL/SQL Exercise 7: Packages

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### Exercise 7: Packages

Scenari- 1: Group all customer-related procedures and functions int- a package.

Question: Create a package CustomerManagement with procedures for adding a new customer, updating customer details, and a function t- get customer balance.

Scenari- 2: Create a package t- manage employee data.

Question: Write a package EmployeeManagement with procedures t- hire new employees, update employee details, and a function t- calculate annual salary.

Scenari- 3: Group all account-related operations int- a package.

Question: Create a package AccountOperations with procedures for opening a new account, closing an account, and a function t- get the total balance of a customer across all accounts.

Schema t- be Created

```
CREATE TABLE Customers (  
  CustomerID NUMBER PRIMARY KEY,  
  Name VARCHAR2(100),  
  DOB DATE,  
  Balance NUMBER,  
  LastModified DATE  
);  
  
CREATE TABLE Accounts (  
  AccountID NUMBER PRIMARY KEY,  
  CustomerID NUMBER,  
  AccountType VARCHAR2(20),  
  Balance NUMBER,  
  LastModified DATE,  
  FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)  
);  
  
CREATE TABLE Transactions (  
  TransactionID NUMBER PRIMARY KEY,  
  AccountID NUMBER,  
  TransactionDate DATE,  
  Amount NUMBER,  
  TransactionType VARCHAR2(10),  
  FOREIGN KEY (AccountID) REFERENCES Accounts(AccountID)  
);  
  
CREATE TABLE Loans (  
  LoanID NUMBER PRIMARY KEY,
```

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```
CustomerID NUMBER,  
LoanAmount NUMBER,  
InterestRate NUMBER,  
StartDate DATE,  
EndDate DATE,  
FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)  
);  
  
CREATE TABLE Employees (  
EmployeeID NUMBER PRIMARY KEY,  
Name VARCHAR2(100),  
Position VARCHAR2(50),  
Salary NUMBER,  
Department VARCHAR2(50),  
HireDate DATE  
);
```

### Example Scripts for Sample Data Insertion

```
INSERT INTO Customers (CustomerID, Name, DOB, Balance, LastModified)  
VALUES (1, 'John Doe', TO_DATE('1985-05-15', 'YYYY-MM-DD'), 1000, SYSDATE);  
INSERT INTO Customers (CustomerID, Name, DOB, Balance, LastModified)  
VALUES (2, 'Jane Smith', TO_DATE('1990-07-20', 'YYYY-MM-DD'), 1500, SYSDATE);  
INSERT INTO Accounts (AccountID, CustomerID, AccountType, Balance, LastModified)  
VALUES (1, 1, 'Savings', 1000, SYSDATE);  
INSERT INTO Accounts (AccountID, CustomerID, AccountType, Balance, LastModified)  
VALUES (2, 2, 'Checking', 1500, SYSDATE);  
INSERT INTO Transactions (TransactionID, AccountID, TransactionDate, Amount, TransactionType)  
VALUES (1, 1, SYSDATE, 200, 'Deposit');  
INSERT INTO Transactions (TransactionID, AccountID, TransactionDate, Amount, TransactionType)  
VALUES (2, 2, SYSDATE, 300, 'Withdrawal');  
INSERT INTO Loans (LoanID, CustomerID, LoanAmount, InterestRate, StartDate, EndDate)  
VALUES (1, 1, 5000, 5, SYSDATE, ADD_MONTHS(SYSDATE, 60));  
INSERT INTO Employees (EmployeeID, Name, Position, Salary, Department, HireDate)  
VALUES (1, 'Alice Johnson', 'Manager', 70000, 'HR', TO_DATE('2015-06-15', 'YYYY-MM-DD'));  
INSERT INTO Employees (EmployeeID, Name, Position, Salary, Department, HireDate)  
VALUES (2, 'Bob Brown', 'Developer', 60000, 'IT', TO_DATE('2017-03-20', 'YYYY-MM-DD'));
```